Name ______________________ **Date** ______________ **Period** ________

Lunar Greenhouse

Campaign 1 · Case 02 · Shackleton Crater, Lunar South Pole

SOLAR AGRICULTURAL AGENCY
Student Mission

MISSION QUESTION

Why do healthy lunar tomato plants form flowers but no fruit?

YOUR ROLE

Process Modeler · Reconstruct the sequence, locate the first failure, then test the diagnosis.

SCIENCE FOCUS · ESTABLISHED + FICTIONAL CONTEXT

Tomato reproduction is a sequence: pollen must be released, reach a receptive stigma, germinate, grow a pollen tube, and complete fertilization before fruit set. The lunar greenhouse and exact readings are fictional case evidence.

TASK 01 1 · Vocabulary

Anther	Structure that produces and holds pollen.
Fertilization	Joining of reproductive cells at an ovule.
Fruit set	Beginning of fruit development.
Pollen tube	Growth from a pollen grain through the style.
Pollination	Transfer of pollen to a receptive stigma.
Stigma	Receptive surface where pollen must arrive.

TASK 02 2 · Initial thinking

A plant is vigorous and flowering but produces no fruit. Which reproductive step would you investigate first, and what evidence would you seek?

TASK 03 3 · Model the pollination sequence

Use each word-bank phrase once. Place the six phrases in process order.

WORD BANK **fruit set · physical agitation · viable pollen in anthers · fertilization · pollen tube growth · pollen reaches stigma**

1	→	2	→	3	→	4	→	5	→	6
PROCESS STEP		PROCESS STEP		PROCESS STEP		PROCESS STEP		PROCESS STEP		PROCESS STEP

PROCESS MODEL Curriculum-original sequence schematic. Read left to right; arrows show dependency.

Model rule: A failed event can interrupt every later event. Do not assume a later structure is defective merely because the process never reaches it.

TASK 04 4 · Identify the failed step

Mark the **first** step that fails. Use two observations to locate it and state one downstream result.

Failed step

Observation 1

Observation 2

Downstream result

TASK 05 5 · Test the competing explanations

Mark **Supported** or **Weakened**, then give one evidence-based reason.

Possible diagnosis	Classification	Evidence-based reason
Lunar regolith is toxic to fruit development.		
The light spectrum lacks ultraviolet light needed for fruiting.		
No effective pollen release and transfer is occurring.		
Carbon dioxide is too high for tomatoes.		

TASK 06 6 · Diagnose and reject an alternative

Diagnosis: state the cause and the failed biological step.

Rejected alternative: name it and give the evidence that weakens it.

TASK 07 7 · Claim-Evidence-Reasoning

Claim

1–2 sentences

Evidence

2–3 specific pieces from more than one source

Reasoning

Explain the mechanism connecting evidence to claim

TASK 08 8 · Design a reliable pollination support

Propose a method that supplies the missing physical action without animal pollinators.

Design

One criterion

One constraint

Mechanism check: restored step and success evidence

TASK 09 9 · Exit ticket

A second sealed habitat grows a different self-pollinating crop. Flowers and viable pollen are present, but no fruit forms. Which step should investigators test first, and what observation would show whether it is failing?